📘 Chapter 3 – Introduction to Linux & Basic Linux Commands

🐧 Introduction to Linux

📘 What is Linux?

Linux is a type of Operating System (OS).

But technically:

🧠 **Linux itself is mainly a Kernel.**

Different companies and developers use the Linux kernel to create their own operating systems called:

🐧 **Linux Distributions (Distros)**

📌 Examples of Linux Distributions

Linux Distribution	**Purpose**
Kali Linux	Ethical Hacking & Penetration Testing
Ubuntu	General Use
Parrot OS	Security & Privacy
Red Hat	Enterprise Servers
Debian	Stable Linux OS

🧩 Daily Life Example

Think of Linux Kernel like:

🚗 A car engine.

Different companies build different cars using the same engine.

Similarly:

Different Linux operating systems are built using the Linux kernel.

🛡️ Why Linux is Important in Ethical Hacking?

Most cybersecurity professionals use Linux because:

✅ Fast
✅ Lightweight
✅ Powerful command line
✅ Better networking tools
✅ Highly customizable
✅ Secure and stable

🖥️ GUI vs CLI (Command Line)

📘 GUI (Graphical User Interface)

GUI means using:

- Mouse
- Icons
- Windows
- Buttons

Example:
Windows Operating System

📘 CLI (Command Line Interface)

CLI means interacting using commands in terminal.

Linux heavily uses CLI.

⚡ Why Command Line is Faster?

The command line consumes fewer system resources compared to GUI.

That is why:

🌐 Most web servers are deployed on Linux servers.

Because servers need:

- Better performance
- Faster processing
- Stability

🧩 Example

A hacker or admin can manage:

- 100 servers
- Network devices
- Security tools

using commands much faster than clicking manually.

💻 Opening Linux Terminal

The terminal is used to execute Linux commands.

In Kali Linux:

You can open terminal using:

- Terminal icon
- Shortcut keys
- Application menu

📂 Linux File System Structure

Linux stores files using a hierarchical directory structure.

Everything starts from:

📁 **/**

This is called:

🌳 **Root Directory**

🪟 Windows vs Linux Path Example

Windows	Linux
C:\Users\Admin\Desktop	/home/admin/Desktop

📌 Important Linux Directories

🏠 1. /home

📘 Purpose

Stores user data and personal files.

🧩 Example

/home/admin

Contains:

- Desktop
- Downloads
- Documents
- Pictures

🧩 Windows Equivalent

C:\Users\Admin

📦 2. /var

📘 Meaning

VAR = Variable Data

This directory stores frequently changing data.

📌 Contains

- Logs
- Cache
- Mail data
- Temporary service files

📜 Important Log Files

🔐 3. Authentication Logs

/var/log/auth.log

Contains:

- Failed login attempts
- Authentication activity
- SSH login attempts

💻 4. System Logs

/var/log/syslog

Contains:

- System events

- Service messages
- General OS logs

🌐 5. Apache Web Server Logs

/var/log/apache2

Contains:

- Website access logs
- Visitor IP addresses
- Error logs

🧩 Example

If you host a website:

All visitor activity gets logged here.

🧠 Why Logs Are Important in Cybersecurity?

Logs help detect:

- Hack attempts
- Failed logins
- Malware activity
- Unauthorized access

Logs are extremely important in:

🕵️ Digital Forensics

⚙️ Kernel Concept

📘 What is Kernel?

Kernel is the core part of the operating system.

It acts as a bridge between:

- Software
- Hardware

🧩 Example

When you move your mouse:

- Application sends request
- Kernel communicates with hardware
- Hardware responds

Kernel manages this interaction.

📌 Kernel Responsibilities

✅ Memory Management
✅ Device Communication
✅ Process Management
✅ Hardware Interaction

📜 Kernel Logs

/var/log/kern.log

Contains:

- Hardware errors
- Driver issues
- Kernel problems

🧩 **Example**

If:

- Keyboard stops working
- Hardware crashes

You can check kernel logs.

⚙️ /etc Directory

📘 **Purpose**

Contains system configuration files.

📌 **Includes**

- Network configuration
- User settings
- Service configuration

🧩 **Example**

/etc/passwd

Contains user account information.

📁 /tmp Directory

📘 **Purpose**

Stores temporary files.

🧩 **Example**

Applications store temporary working data here.

⚠️ **Security Note**

Attackers and malware sometimes hide temporary files in /tmp.

📦 /bin Directory

📘 **Purpose**

Contains essential Linux commands and binaries.

📌 **Examples**

/bin/ls
/bin/cp
/bin/mv

These are basic commands required by all users.

🔐 /sbin Directory

📘 **Meaning**

SBIN = System Binary

Contains administrative and system management commands.

📌 **Examples**

- Firewall tools
- Network tools
- System utilities

Mostly used by:

👑 **Root User**

📂 /usr Directory

📘 Purpose

Contains installed applications and user-related software.

🧩 Example

Applications installed by users are commonly stored here.

📂 /opt Directory

📘 Purpose

Stores third-party software and optional applications.

🧩 Example

Google Chrome or custom software may install files here.

🚀 /boot Directory

📘 Purpose

Contains files required for system startup.

📌 Important Boot Loader

🔥 GRUB Loader

GRUB helps load the operating system during boot.

🧩 Example

When computer starts:

BIOS/UEFI → GRUB → Linux Kernel

💽 /dev Directory

📘 Purpose

Contains device files.

📌 Examples

- Hard Disk
- USB
- Keyboard
- Mouse

Everything in Linux is treated like a file.

📁 /proc Directory

📘 Purpose

Contains process and system information.

Acts somewhat like:

📊 Task Manager

in Windows.

📌 Example

/proc/meminfo

Shows RAM/memory information.

👤 Linux Users

💲 Normal User

When terminal shows:

$

It means:

✅ Logged in as normal user.

Normal users have limited permissions.

👑 Root User

Root user is similar to:

🛡️ Administrator in Windows

Root has full system access.

🔥 Switching to Root User

Command

```
sudo su
```

📘 Meaning

Command	Meaning
sudo	Super User Do
su	Switch User

⚠️ Important Note

Root access is powerful and dangerous.

Wrong commands can damage the entire system.

🧭 Basic Linux Commands

📂 1. ls Command

📘 Purpose

Lists files and folders.

Command

ls

🎨 Color Meanings in Linux

Color	Meaning
🔵 Blue	Folder
⚪ White	Text File
🟣 Purple	Media Files

🧩 Example

ls

Output may show:

Documents song.mp3 notes.txt

📍 2. pwd Command

📘 Meaning

PWD = Print Working Directory

Shows current folder location.

Command

pwd

🧩 Example

/home/admin/Desktop

🧹 3. clear Command

📘 Purpose

Clears terminal screen.

Command

clear

📋 4. ls -l Command

📘 Purpose

Shows detailed file information.

Command

ls -l

📌 **Displays**

- Permissions
- Owner
- File size
- Date
- File type

👀 5. ls -a Command

📘 **Purpose**

Shows hidden files.

Command

ls -a

📌 **Hidden Files**

Linux hidden files start with:

.

Example:

.bashrc

📁 6. cd Command

📘 **Meaning**

CD = Change Directory

Used to move between folders.

Command

cd Documents

⚠️ **Linux is Case Sensitive**

Example:

cd Documents

✅ Correct

cd documents

❌ Wrong (if folder name starts with capital D)

📂 **7. Going Back Directories**

One Folder Back

cd ..

Two Folders Back

cd ../..

📁 **8. mkdir Command**

📘 **Meaning**

MKDIR = Make Directory

Creates a new folder.

Command

mkdir hacking_notes

📄 9. touch Command

📘 Purpose

Creates empty files.

Command

touch notes.txt

📖 10. cat Command

📘 Purpose

Displays file content.

Command

cat notes.txt

📌 Example Output

Linux is important in cybersecurity.

📦 11. cp Command

📘 Meaning

CP = Copy

Copies files or folders.

Command

cp notes.txt backup.txt

🚚 12. mv Command

📘 Meaning

MV = Move

Used to:

- Move files
- Rename files

📌 Move File Example

mv notes.txt /home/admin/Documents

📌 Rename File Example

mv old.txt new.txt

❌ 13. rm Command

📘 Meaning

RM = Remove

Deletes files.

Command

rm notes.txt

☠️ 14. Force Delete

Command

rm -rf foldername

⚠️ Meaning

Option	Meaning
-r	Recursive
-f	Force

🚨 Dangerous Command Warning

rm -rf can permanently delete files/folders.

There is usually:

❌ No recycle bin recovery.

🛡️ Kali Linux

📘 Definition

Kali Linux is a Linux distribution specially made for:

- Ethical Hacking
- Penetration Testing
- Digital Forensics
- Security Research

📦 Kali Linux Includes Tools Like

- Nmap
- Hydra
- Burp Suite
- Metasploit
- Aircrack-ng

🧠 Quick Revision Table

Topic	Key Point
Linux	Open-source operating system
Kernel	Bridge between software & hardware
/var	Logs & variable data
/etc	Configuration files
/boot	Boot files
Root User	Full privileges
ls	List files
pwd	Show current path
cd	Change directory
mkdir	Create folder
rm	Delete file
Kali Linux	Cybersecurity Linux distro

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